



1125

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : **Nakshatra**

Team Leader Name : **Avinash Chandra Gupta**

Problem Statement : Detection and Characterization of Subsurface Ice in Lunar South Polar Regions Using Chandrayaan-2 Radar and Imagery Data for Landing Site and Rover Traverse Planning

TEAM MEMBERS

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Difference from existing ideas

Slope-Corrected Polarimetry vs. Flat Thresholding: Tilted terrain alters local radar incidence angles, causing rocky rims to mimic ice. We dynamically normalize *CPR* and *DOP* against local facet slopes from the 5m DEM to eliminate false positives.

Subsurface 1D Heat-Diffusion vs. Skin-Temp Masks: Diviner $T_{\max}$ only measures surface skin temperature. We run a 1D thermal model using regolith thermal inertia ($I \approx 40 \text{ J m}^{-2} \text{ K}^{-1} \text{ s}^{-1/2}$) to locate stable ice insulated at L-band depths (1-3 m) beneath warm ($>110 \text{ K}$) surface soils.

Dual-Band Phase Decoherence Mapping: We measure the active spatial phase delta ($\Delta\Phi = \Phi_L - \Phi_S$) to prove a stratified depth model (surface frost vs. deep reserves) instead of a flat 2D binary mask.

Solving the Problem

Fuses active L/S DFSAR data with topographic grids to programmatically export 3 flight-ready deliverables:

3D Volumetric Reserve Map: Correlates S-band and L-band skin-depth models to calculate physical water-ice mass estimates.

Geodetically Corrected Touchdown Coordinates: Ultra-precise landing coordinates corrected for polar stereographic projection warping (pyproj ellipsoid transform).

Soil Cohesion & Trafficability Cost Map: Translates radar dielectric constants (T-Ratio) and thermal gradients into wheel-slip hazards to prevent rover wheel-sinkage in fluffy, low-density PSR regolith.



• Conclusive simulation to physics chain

Topography Normalized Radar



Subsurface Heat Diffusion



Regolith Trafficability

• **Deterministic flight pipeline:** The entire pipeline executes programmatically in a single automated run, translating raw active spacecraft radar files into a flight-computer-ready rover navigation vector.

• **Topographic Power & Comm Horizons:** Our pathfinder calculates geometric line-of-sight to Earth (communication) and blocks paths entering solar shadows relative to the ultra-low $\approx 1.54^\circ$ polar sun.

Multi-Sensor Polarimetric Data Fusion

- **Active DFSAR Processing:** Ingests raw, co-registered S/L-band radar simultaneously to extract active polarimetric profiles.
- **Auto-Co-registration:** Standardizes varying sensor spatial resolutions (5 m and 20 m DEMs) onto a unified projection grid.
- **No-Data Grid Cleaning:** Interpolates high-altitude sensor clipping and void artifacts to eliminate calculation crashes.

Depth-Discriminated Ice Characterization

- **Stratified Volatile Profiling:** Profiles S-band (12.5 cm surface frost) vs. L-band (1-3\text m deep stable ice) instead of flat binary masks.
- **Rock Scattering Filter:** Excludes warm rocky slopes by pairing CPR > 1.0 with DOP > 0.5 volume scattering thresholds.
- **Thermal Stability Validation:** Intersects radar assets with Diviner maps using a strict maximum temperature cutoff ($T_{\max} < 110$ K).

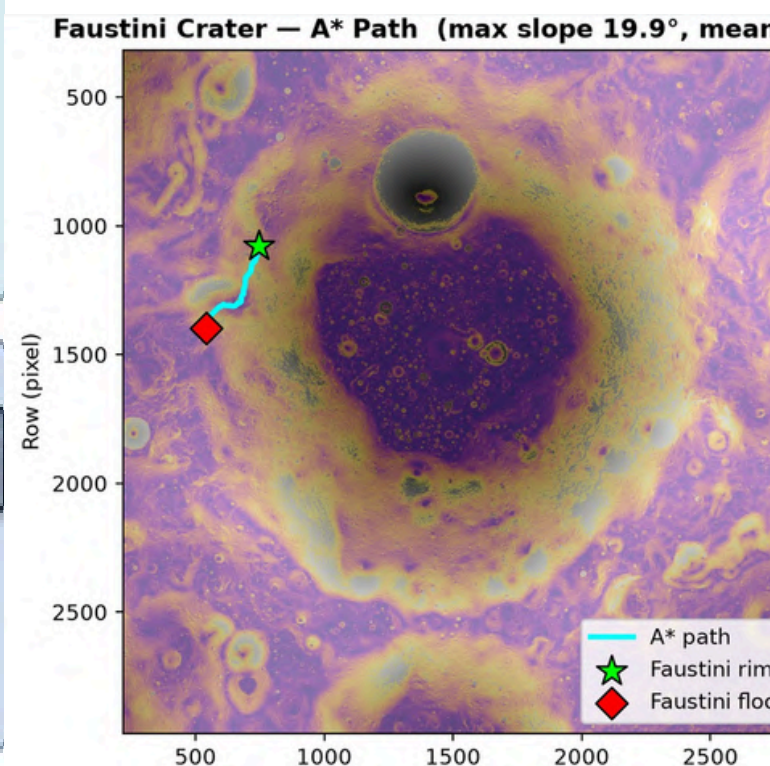
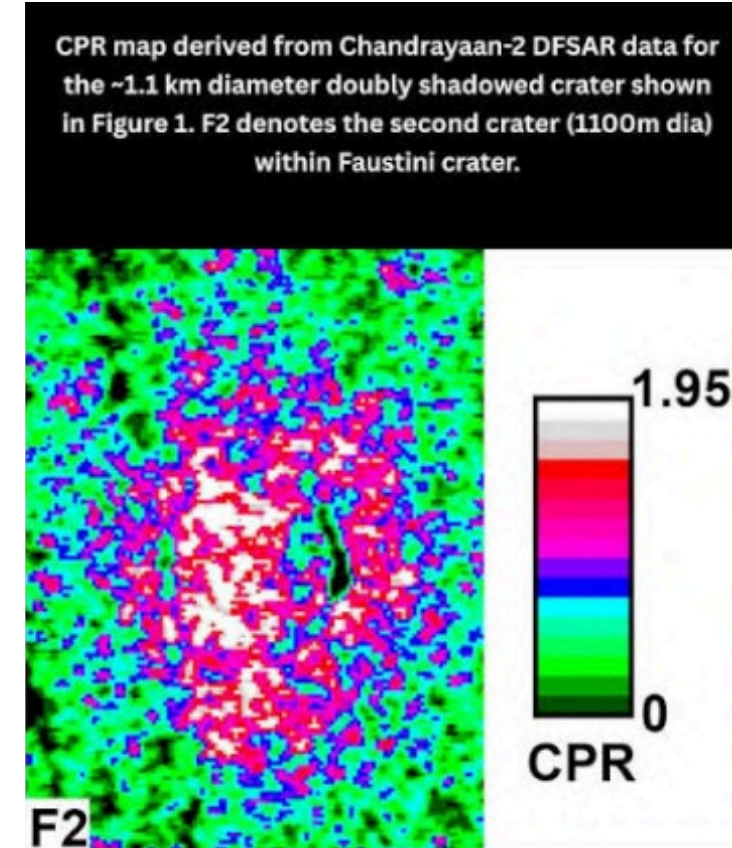
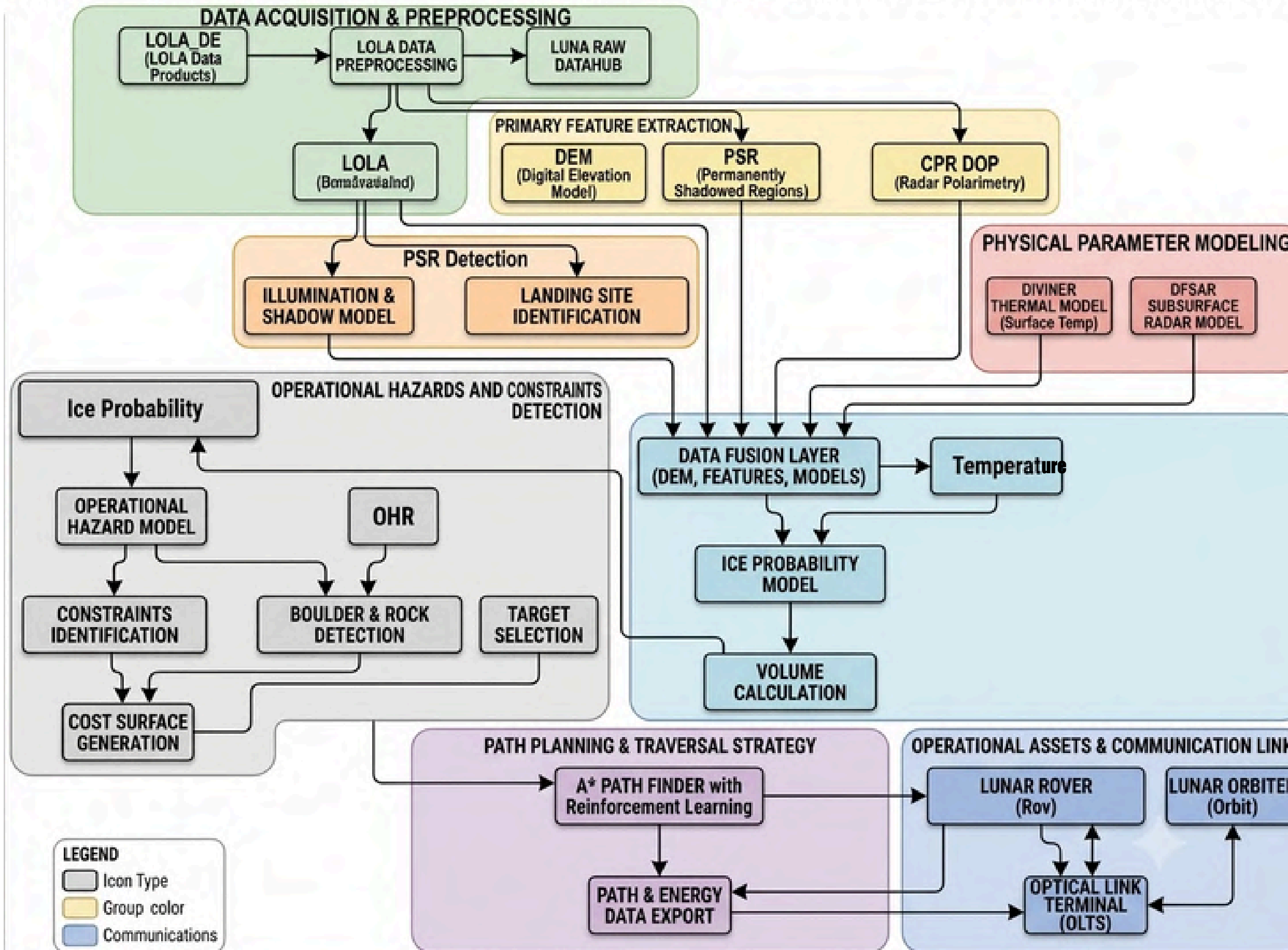
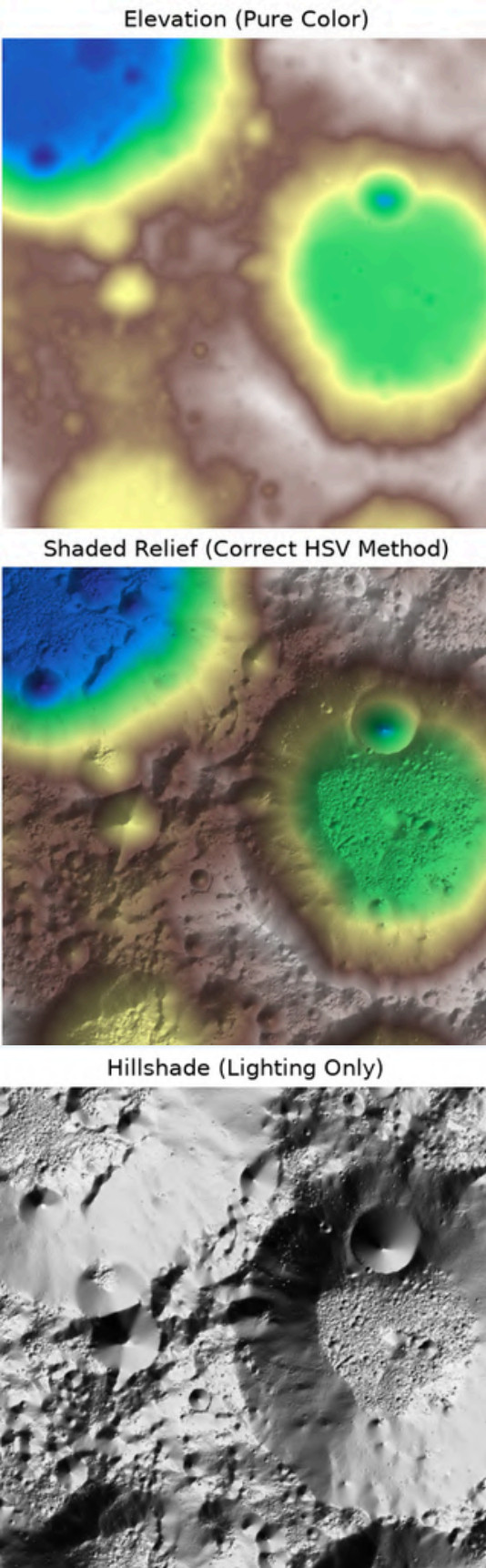
Dual-Pass Polar Ray-Tracer

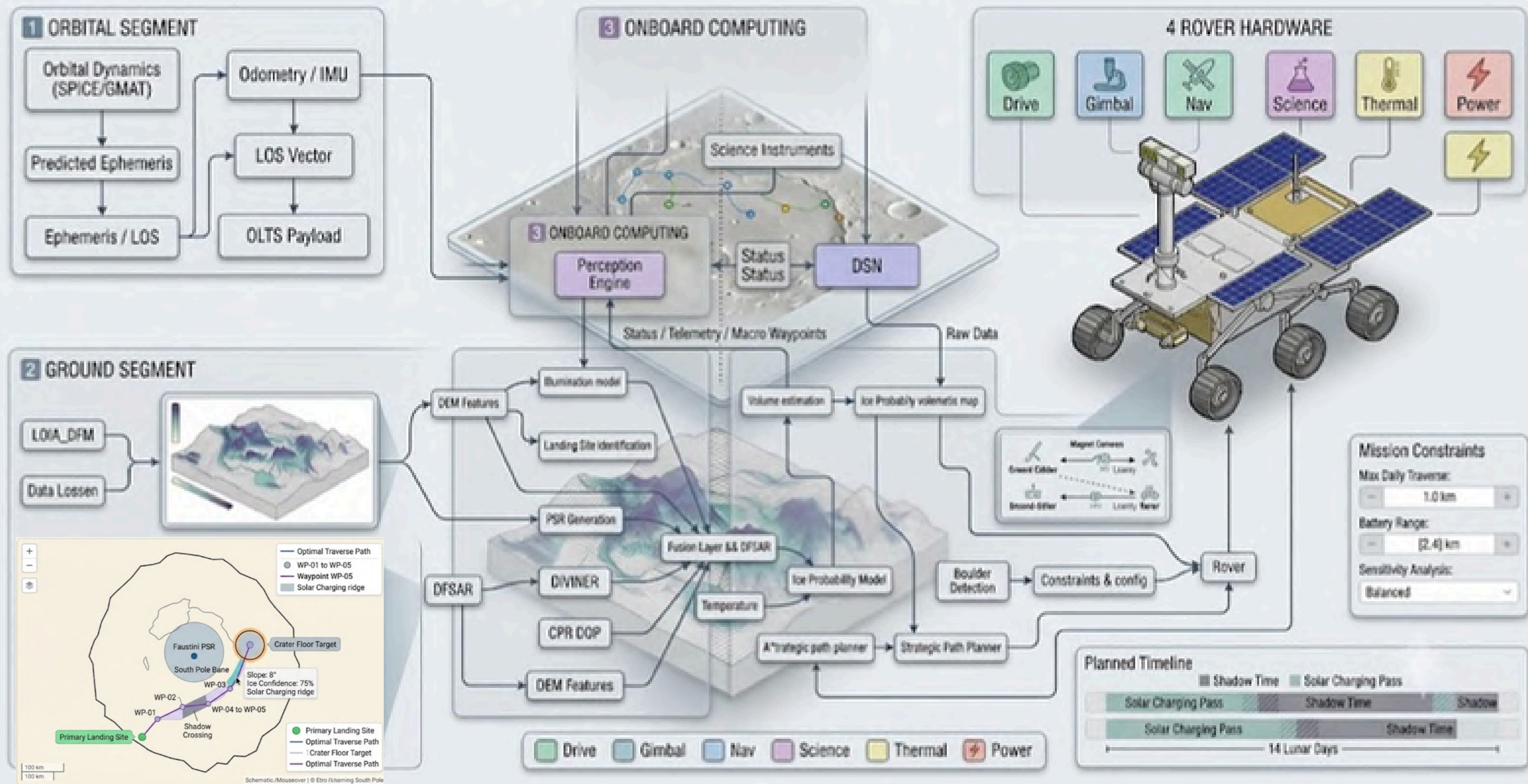
- **Pass 1 (Direct Obstruction):** Simulates low-angle parallel solar rays ($\approx 1.54^\circ$) over local topography to outline macro-shadowed cold traps (PSRs).
- **Pass 2 (Secondary Radiative Shielding):** Calculates geometric view factors from surrounding topography, isolating Doubly Shadowed Regions (DSRs) shielded from secondary scattered thermal emissions.

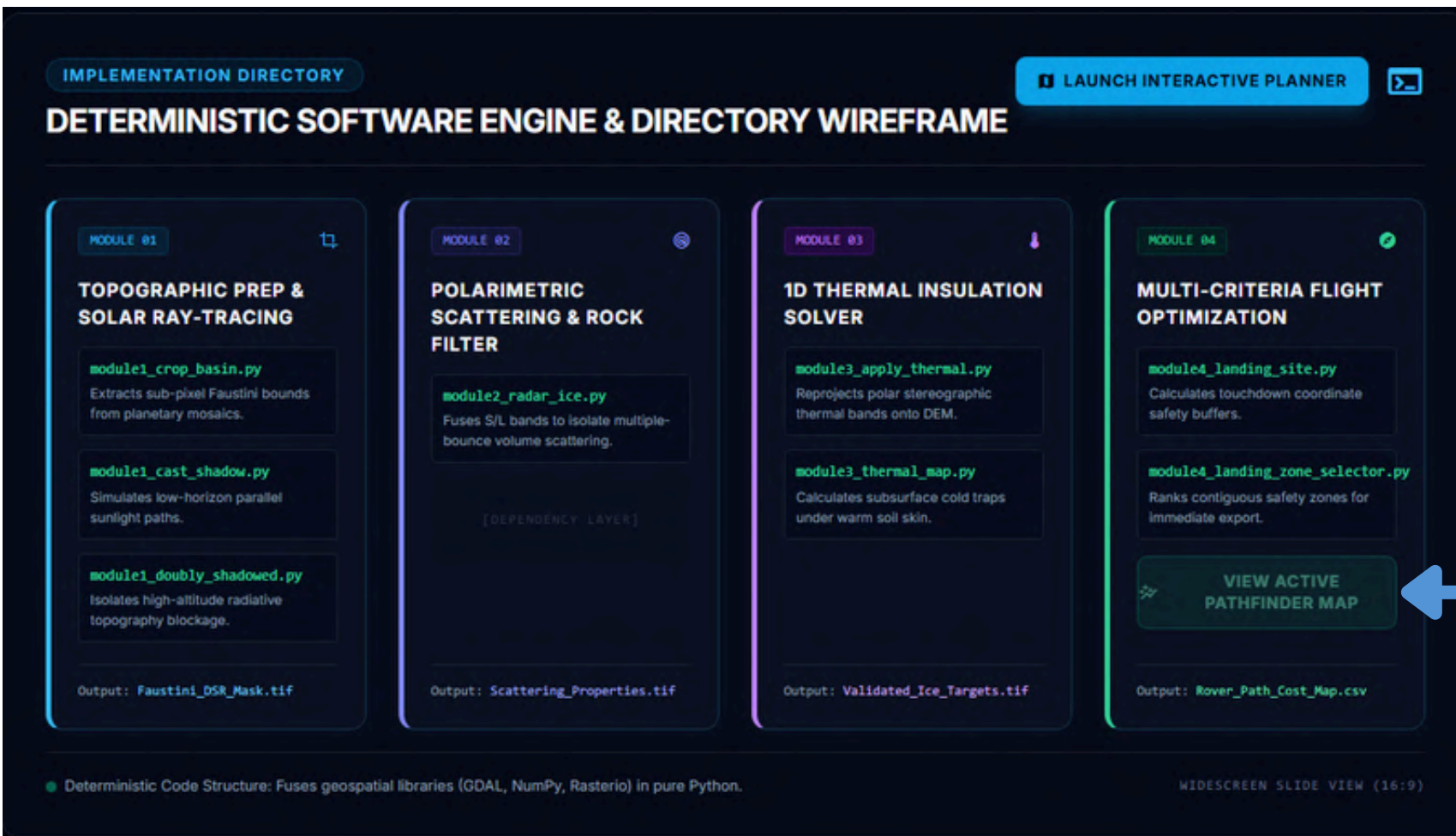
Lander & Traverse Optimizer

- **Terrain Safety Screening:** Masks out hazardous terrain by enforcing strict landing slope limits ($\approx 15^\circ$) over local LOLA grids.
- **Proximity Buffering:** Enforces a safe 500-2000 m standoff window to protect volatile samples from rocket engine plume contamination.
- **Continuous Cost Output:** Auto-ranks touchdown zones and exports terrain matrices directly as float arrays for A* and RL flight computer routing.

PROCESS FLOW DIAGRAM



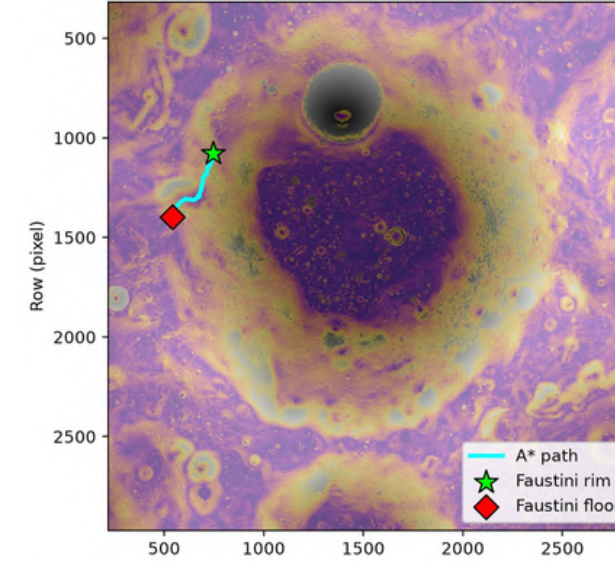




Layer control for Masks

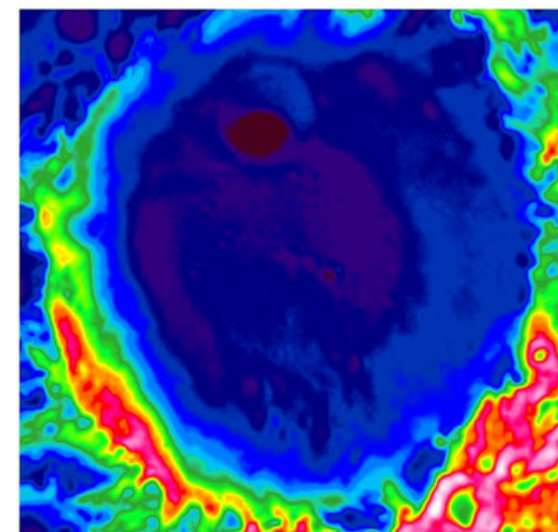
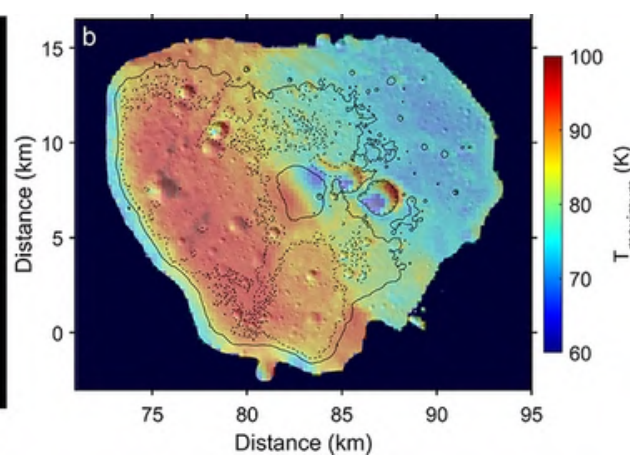
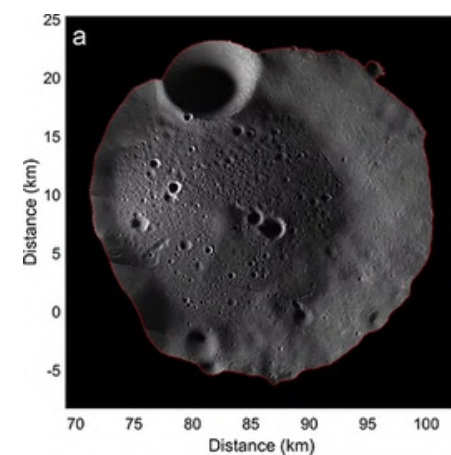
Rover Monitoring Tab

Faustini Crater — A* Path (max slope 19.9°, mean 10.9°)



Interactive Rover Plan Map

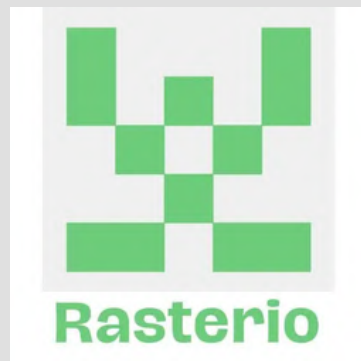
Module Codebase



Temperatrue mask



Geospatial Engineering



Scientific Computing



Spatial Analysis



Visualisation and Verification



Component	Purpose	Estimated Cost (₹ Crore)
Dual-band L/S-band Ground Penetrating Radar	Detection of subsurface ice and depth profiling	100
Radiation-hardened onboard computer	AI inference, ice volume estimation, and real-time path planning	25
Navigation sensors (stereo cameras, IMU, LiDAR)	Terrain perception and rover localization	40
Software & AI development	Ice estimation model, dynamic cost mapping, and path optimization	30
System integration & environmental testing	Hardware integration, thermal-vacuum, vibration, and radiation testing	60



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